

Manufacturing process

Week -6

1. Estimation of Length of Raw material required to convert Circular rod to Hexagon and Calculate the Production Cost considering direct and Indirect expenses.
2. Explain the working principle of Power hammer
3. Show the you tube videos on components produced by Power forging

Power Hammer

Hand hammer blows impact will not be always sufficient enough to affect the proper plastic flow in a medium sized or heavy forging. It also causes fatigue to the hammer man. To have heavy impact or blow for more plastic deformation, power hammer are generally employed.

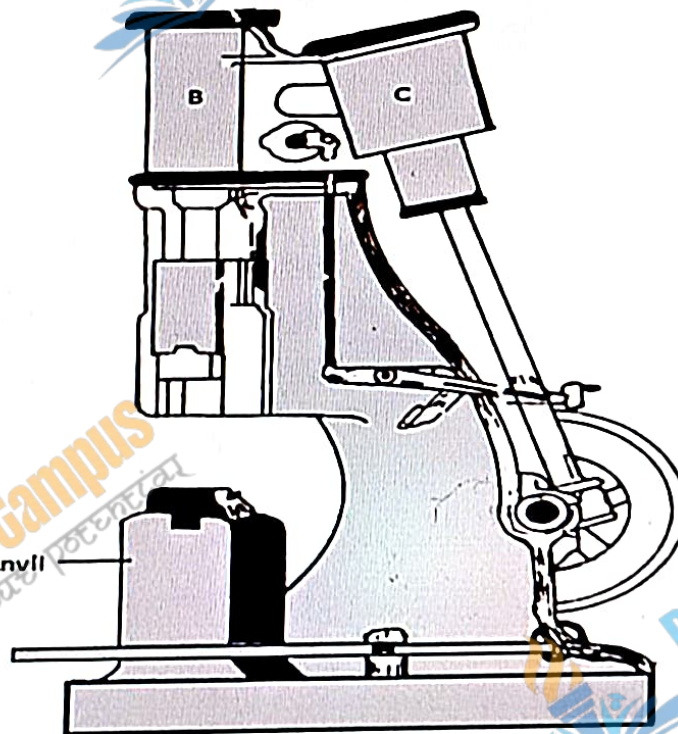
These hammers are operated by compressed air, steam, oil pressure, spring and gravity.

Types of Power Hammers

There are various different types of forging power hammers that are widely used

1. Pneumatic Hammers
2. Steam or Air Hammers
3. Hydraulic Power Hammer
4. Lever-Spring Hammers

1. Pneumatic Power Hammer



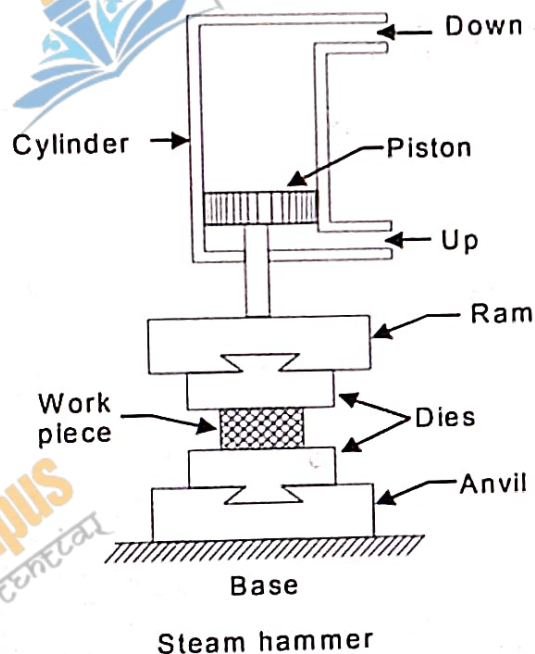
Construction & Working

Pneumatic power hammers are quite a typical form of mechanical device which is responsible for carrying the cylinder in which the piston works. This piston is found to be connected to the main motor shaft with the help of a crank and a connecting rod mechanism. A hand lever is also involved which operates the air valve.

The air valve is provided through the different cylinders. Another piston also works inside the cylinder which carries the tup at its bottom and is made to slide fixed guides. In order to make the tup move, the piston in the cylinder is moved downwards in order to create the vacuum above the piston inside the other cylinder. This is responsible for providing a suction effect on the piston due to which the tup starts lifting.

Near the end of an upward stroke of the piston, in-cylinder, the air is forced from one cylinder to another where the air is compressed due to the upward stroke of the piston. This is responsible for forcing the piston and thus moving the tup downwards with high velocity. This operation is repeated on a regular basis and the required blows are obtained.

2. Steam or Air Power Hammer



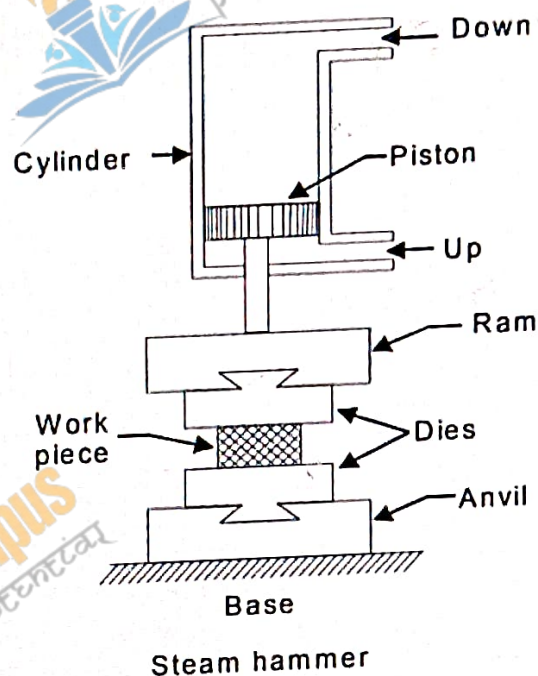
Construction & Working

Pneumatic power hammers are quite a typical form of mechanical device which is responsible for carrying the cylinder in which the piston works. This piston is found to be connected to the main motor shaft with the help of a crank and a connecting rod mechanism. A hand lever is also involved which operates the air valve.

The air valve is provided through the different cylinders. Another piston also works inside the cylinder which carries the tup at its bottom and is made to slide fixed guides. In order to make the tup move, the piston in the cylinder is moved downwards in order to create the vacuum above the piston inside the other cylinder. This is responsible for providing a suction effect on the piston due to which the tup starts lifting.

Near the end of an upward stroke of the piston, in-cylinder, the air is forced from one cylinder to another where the air is compressed due to the upward stroke of the piston. This is responsible for forcing the piston and thus moving the tup downwards with high velocity. This operation is repeated on a regular basis and the required blows are obtained.

2. Steam or Air Power Hammer

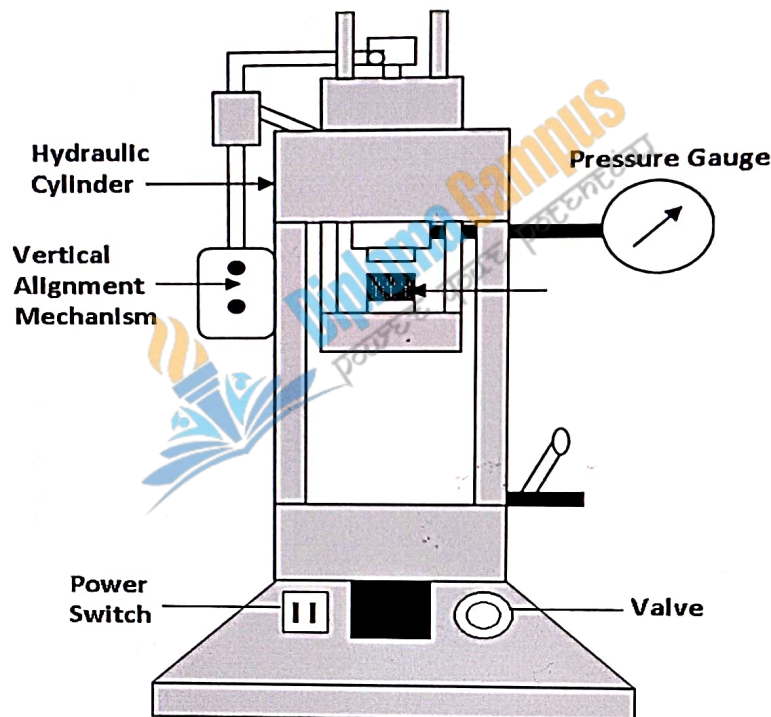


Construction & Working

A steam-power hammer is referred to as those which can be performed by the effect of either steam or compressed air. These are the device which includes a moving ram, rod, piston, a lifting tool, a double-acting high-pressure steam cylinder, housing or frame, and anvil. These are the forging power hammers wherein the steam gets entered initially and then at the bottom of the cylinder; the piston gets lifted upwards along with the other moving parts. The amount of force is considered as 3m/sec whereas the mass of the moving parts is nearly 5000kg.

Whenever the blow is required, the lever gets energized automatically where the top sliding valve is opened in order to receive the steam from above of the cylinder and the exhaust steam which remains on the bottom. Thus, the required blows are obtained.

3. Hydraulic Power Hammer



Hydraulic Press

Construction & Working

The hydraulic hammers also known as hydraulic presses. It uses 200 to 300 times higher oil pressure than the atmospheric pressure in the hydraulic cylinder. Hydraulic presses are applied in heavy forgings.

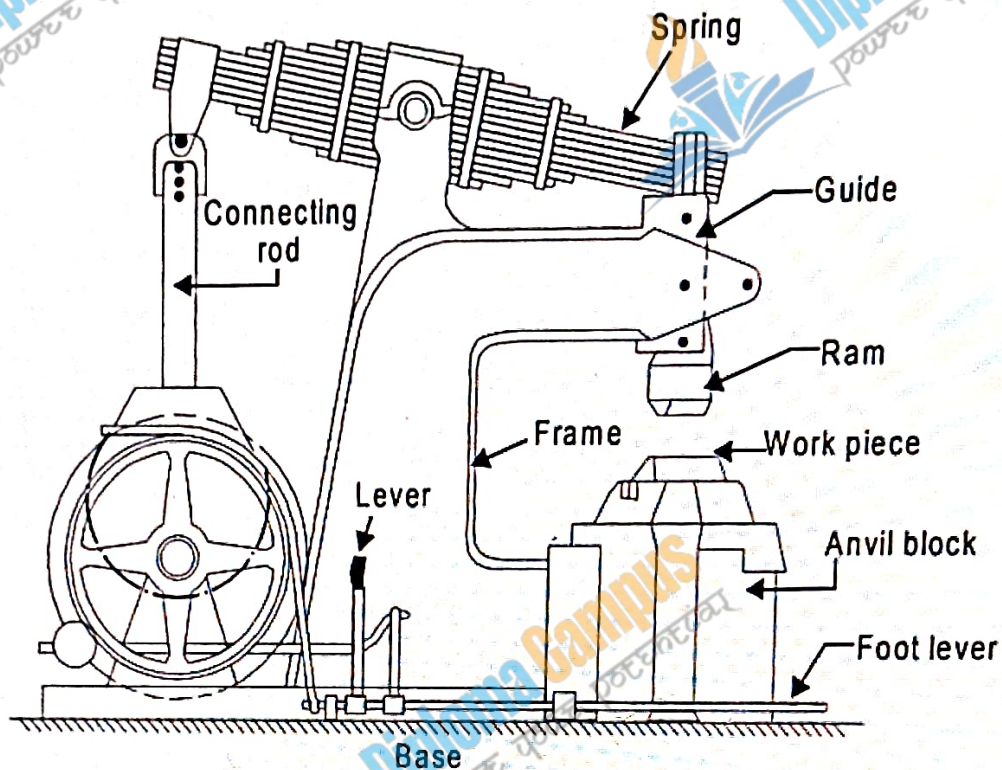
It consists of a press and hydraulic drive. The rate of production in hydraulic presses is normally faster than other hammer forgings, as the entire operation is completed in a single squeezing action.

The hydraulic hammer has low noise and vibration. In these presses, the speed, blow pressure and die travel are automatically controlled.

Presses that can provide a force of 75000 tons are pretty common. Hydraulic presses are used in forging industries almost ranges from 1000 tons to 10,000 tons. These presses are used for the production of nuts, bolts, screws, and rivets.



4. Lever Spring Hammer



Construction & Working

A lever-spring hammer is a quite light type of power hammer that is mostly used for small forgings. These types of hammers consist of a heavy frame with a vertical projection at its top which acts as a housing to bearing in which the laminated spring is found oscillating. There are two ends of this spring wherein one end is responsible for carrying a connecting rod and the other end a vertical tup. The tup is responsible for carrying the weight and moves vertically upwards and downwards within the fixed guides.

The connecting rod is attached with an eccentric sheave at its lower end. Further, the eccentric sheave is connected to the crank wheel which is responsible for operating the hammer and pressing the treadle downwards.

This is what makes the sheave rotate through the crank wheel and therefore the laminated spring starts oscillating in the bearing. This continuous oscillation of the spring leads the tup to move upwards and downwards. Therefore, the required blows are provided on the job and the hand lever is operated in order to adjust the stroke of the connecting rod and the intensity of blows.

AIM:

To make a hexagonal bolt head from a round bar.

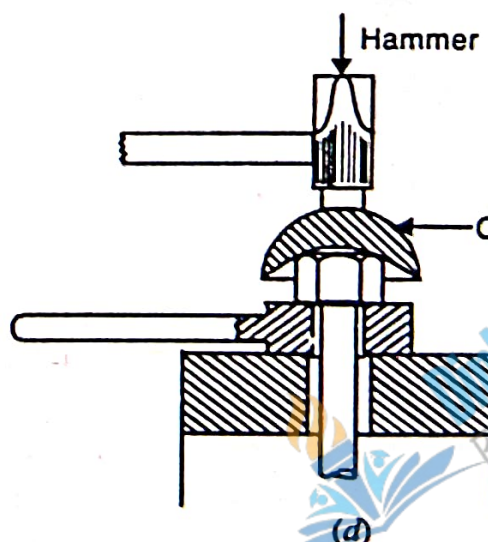
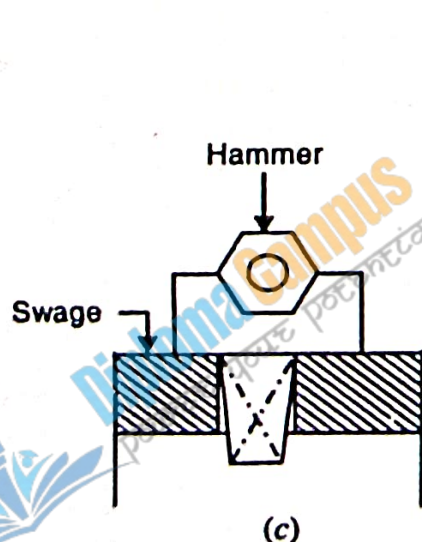
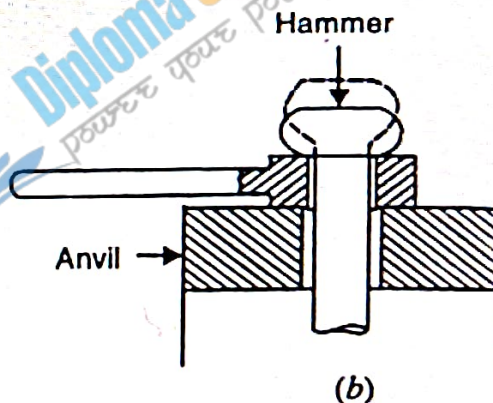
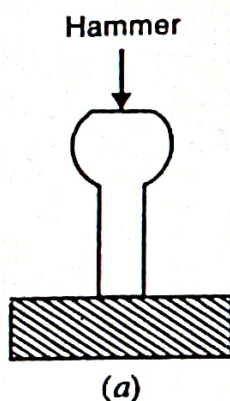
Tools Required:

The tools required are anvil, hand hammer, tong, bush or size plate, flattener swage, cupping tool, scale and caliper.

Procedure

The sequence of operations are as follows:

1. First of all, one end of the bar is heated up for a sufficient length to make the head.
2. The heated end is jumped up on the anvil, as shown in Fig. (a).
3. The head is flattened by hammering against the end of a bush through which the shank passes, as shown in Fig. (b).
4. The head is formed to hexagonal shape by keeping it on the bottom swage, as shown in Fig. (c).
5. The camber is given on the head by means of a cupping tool using the bush as support as shown in Fig. (d).



Visual Watermark